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Introducing the READY Study: DHH Young people's Well-Being and Self-Determination

Alys Young[1,2,*], Francisco Espinoza[1], Claire Dodds[1], Garry Squires[3], Katherine Rogers[1], Helen Chilton[4] and Rachel O'Neill[5]

[1]SORD (Social Research with Deaf people), University of Manchester, UK
[2]Centre for Deaf Studies, University of the Witwatersrand, South Africa
[3]SEED (School of Education, Environment and Development), University of Manchester, UK
[4]MANCAD (Manchester Centre for Audiology and Deafness), University of Manchester, UK
[5]Moray House, School of Education and Sport, University of Edinburgh, UK

*Correspondence should be addressed to Alys Young, SORD, School of Health Sciences, University of Manchester, Jean McFarland Building, Oxford Road, Manchester M13 9PL, UK. E-mail: alys.young@manchester.ac.uk

Abstract

READY is a self-report prospective longitudinal study of deaf and hard of hearing (DHH) young people aged 16 to 19 years on entry. Its overarching aim is to explore the risk and protective factors for successful transition to adulthood. This article introduces the cohort of 163 DHH young people, background characteristics and study design. Focusing on self-determination and subjective well-being only, those who completed the assessments in written English ($n = 133$) score significantly lower than general population comparators. Sociodemographic variables explain very little of the variance in well-being scores; higher levels of self-determination are a predictor of higher levels of well-being, outweighing the influence of any background characteristics. Although women and those who are LGBTQ+ have statistically significantly lower well-being scores, these aspects of their identity are not predictive risk factors. These results add to the case for self-determination interventions to support better well-being amongst DHH young people.

The Recording Emerging Adulthood in Deaf Youth (READY) study is a prospective longitudinal study of deaf and hard of hearing (DHH) youth in Great Britain. It focuses on the transition to young adulthood, conventionally defined as the period from post-16 education or late adolescence into mid-20s. Contemporary DHH young people entering adulthood are different from previous generations (Byatt, Duncan, Dally, & de Beuzeville, 2021). The vast majority will have experienced early hearing detection and intervention (JCIH, 2019), the developmental advantages of which for linguistic, cognitive, and socioemotional development far exceed those of previous generations (Yoshinaga-Itano, 2004; Yoshinaga-Itano, Sedey, Coulter, & Mehl, 1998). This generation has also witnessed the greater recognition, on an international basis, of signed languages and the associated legal and social rights attributed to their users (Wheatley & Pabsch, 2012) along with a growth in public recognition and positive attitudes to signed languages within mainstream media and society (Lawton, 2021).

Improved hearing technologies now offer more access to sound and spoken language in ways that can be personalized to the individual's preferences and strengths (Archbold & Mayer, 2012) that were not available to previous generations growing up. In countries of the developed world, the vast majority of DHH children are educated in mainstream classrooms (Byatt et al., 2021), many without any specifically tailored additional learning support (De Raeve & Lichtert, 2012; Knoors & Marschark, 2012). More broadly, this is the first generation to "grow up digital", sometimes referred to as Generation Z, with easy access to online communication in written and visual forms (Gentina & Parry, 2021; Snap Group, 2019), including signed languages. They are also a potentially prey to the negative effects on mental health increasingly associated with young people's digital lives (Frith, 2017).

Although previous studies have focused prospectively on transition to early adulthood for DHH young people, such as the National Longitudinal Transition Study (NLTS) in the USA (Wagner, Newman, Cameto, & Levine, 2005), these data are up to 20 years old (Palmer, Newman, Davidson, & Cawthon, 2020). NLTS does not capture the impact and effects of the innovations in hearing, sign language, education, and digital contemporary society of Generation Z. Also, NLTS2 did not clearly distinguish the sample by language use, or degree of deafness, using instead dichotomous variables such as "clarity of speech" and "uses sign language" (Johnson, Newman, Cawthon, & Javitz, 2022). Participants in the longitudinal birth cohort studies of early identified deaf children such as the LOCHI study (Ching, Leigh, & Dillon, 2013) and the Wessex study (Kennedy et al., 2006) are only just reaching late adolescence, with very little data on post-16 psychosocial outcomes published. In the Deaf Studies field there is no longitudinal, prospective study that is exclusively self-report by deaf young people (rather than by teachers or parents), similar to those common in general population youth cohort studies such as the Next Steps study in England (Next Steps, n.d.). Although retrospective, secondary data analysis of key outcomes over time

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such as employment (Garberoglio, Palmer, Cawthon, & Sales, 2019) and educational attainment (Newman et al., 2011; O'Neill, Arendt, & Marschark, 2014; Young, Squires, Oram, Sutherland, & Hartley, 2015) is available, real-time tracking of the influence of DHH young people's profiles on the choices, challenges, and outcomes they face in transition to adulthood is lacking.

The READY study was set up to fill this knowledge gap in being an entirely youth self-report study in which participants are followed up in real time over 5 years from first joining the study aged between 16 and 19 years. Its overarching research question is: What are the risk and protective factors relevant to DHH young people's achievement of: autonomy, social development, personal well-being, educational and occupational attainment? These outcomes were chosen as markers of maturing adulthood in line with large-scale population youth studies in the UK (e.g., Millennium Cohort Study, n.d). In addition to annually collecting data on personal, familial, educational, and social profiles of young people, the design focuses on measurement and tracking of three outcome variables relevant to this time of transition and previously recognized as problematic for DHH young people, in addition to educational attainment and employment status.

The first outcome measure is subjective well-being, chosen because individuals with high subjective well-being are reported as being at a lower risk of a variety of psychological and social problems, including depression and maladaptive relationships with others (Bücker, Nuraydin, Simonsmeier, Schneider, & Luhmann, 2018; Park, 2004) particularly in young adulthood. Although the disproportionately high incidence of mental ill health experienced by DHH young people in middle adolescence/early adulthood is well recognized (Fellinger, Hozlinger, Sattel, & Laucht, 2008; Hintermair, 2007; Kouwenberg, Rieffe, & Theunissen, 2011; Roberts et al., 2015; Stevenson, Kreppner, Pimperton, Worsfold, & Kennedy, 2015; Van Gent, Goedhart, Hindley, & Treffers, 2007; Van Gent, Goedhart, & Treffers, 2012), subjective well-being is distinct. It is a subclinical construct generally considered to have three components (Steptoe, Deaton, & Stone, 2015): (i) life evaluation, also referred to as life satisfaction, which concerns an individual's evaluations of their life and their satisfaction with its quality and how good they feel about it; (ii) hedonic well-being, which refers to everyday feelings or moods and focuses on affective components (feeling happy); (iii) eudaimonic well-being, which emphasizes action, agency and self-actualization (e.g., sense of control, personal growth, feelings of purpose, and belonging) and includes judgments about the meaning of one's life (Stewart-Brown et al., 2009). It operates as a protective and enabling trait crucial to positive mental health especially with regard to young people's transitions to adulthood (Department for Education, 2022).

Far less is known about subjective well-being amongst deaf young people rather than mental illness. However, evidence from disability-based inequities in youth subjective well-being points to the influence of social disadvantage and exposure to discrimination as significant (Savage, McConnell, Emerson, & Llewellyn, 2014). Our interest is in identifying which factors might support positive well-being and which might make deaf young people particularly vulnerable to poorer well-being during this period of transition to adulthood.

The second outcome measure in the READY study is self-determination which encompasses the skills associated with being a causal agent in one's life, including decision making, problem solving, self-advocacy, self-management, goal setting, and attainment alongside self- awareness and self-knowledge (Shogren et al., 2015a; Shogren, Wehmeyer, Palmer, Rifenbark, & Little, 2015b). This construct building on Causal Agency Theory (Shogren et al., 2015a) takes account of how opportunities and affordances in one's environment and social, interactive contexts are significant in the development of self-determination. Consequently, the partially accessible or communicatively restrictive environments experienced by some deaf children and young people are germane to the development of self-determination (Garberoglio et al., 2022).

Research with populations of children and young people with disabilities has established a strong relationship between enhanced self-determination and in-school academic performance (Shogren et al., 2015b; Shogren et al., 2017) as well as being a predictor of post-school education, employment outcomes, and community participation (Shogren et al., 2015b; Test et al., 2009). Recent work on self-determination in a Deaf ASL population has shown scores on the Self-Determination Inventory: Student Report more similar to youth without disabilities than youth with disabilities (Garberoglio et al., 2022). As a dispositional trait it is also one open to modification through late adolescence/early adulthood and the focus of specific interventions and curricula in relation to deaf young people (Shogren, Raley, Burke, & Wehmeyer, 2019). Our interest is in gauging what factors influence the self-determination of deaf young people in the READY cohort, and how might self-determination interact with other outcome measures such as employment, educational attainment, and subjective well-being?

The third outcome variable is health. The social determinants of health are a well-recognized framework for understanding the socioeconomic and structural mechanisms leading to less-than-optimal health outcomes for those facing multiple social disadvantages (Marmot, 2020). Perhaps less well recognized, is that poor health is an identifiable risk factor for a range of socioeconomic and psychosocial outcomes in adult life (King's Fund, 2020). Within a prospective longitudinal study design, we were interested to know the starting health status of young people and to track over time the associations of this with desirable outcomes in young adulthood. Although data exist on the lower health status of DHH adults in comparison with the hearing population, including barriers to health access and health literacy, and impacts on health of adverse communication experiences in childhood (Barnett et al., 2011; Emond et al., 2015; Fellinger, Holzinger, & Pollard, 2012; Kushalnagar, Ryan, Paludneviciene, Spellun, & Gulati, 2020), assessment data on health-related quality of life in early adulthood is absent.

In what follows, we outline the study design, focus on the characteristics of the recruited baseline cohort of 163 and present an initial analysis of the data on the outcome measures concerning subjective well-being and self-determination. Health-related quality of life is the subject of a separate publication.

Method

Design

READY is a prospective longitudinal cross-sectional, mixed methods design consisting of annual data collection via on online survey, three standardized assessment instruments and qualitative interviews with a subsample of the recruited cohort.

Research Questions for This Segment of Data

Within the overarching research aim to explore the risk and protective factors in DHH young people's transition to adulthood, in this article we address the following questions:

RQ1: What are the baseline characteristics of the READY cohort?

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RQ2: How do participants score on the outcome measures of subjective well-being and self-determination in comparison with general population data?
RQ3: What are the relative contributions of socio-demographic characteristics to participants' outcome scores on these measures?
RQ4: To what extent does self-determination influence subjective well-being?

Interest in the potential association between self-determination and subjective well-being stems from two considerations. First, from a theoretical perspective, some aspects of subjective well-being overlap with the construct of self-determination; in particular, the eudaimonic aspects of well-being which emphasize action, agency, and self-actualization (Stewart-Brown et al., 2009) and the agentic action domain of the self-determination construct. Second, self-determination is clearly recognized as a modifiable dispositional trait that can be improved through deliberate programs of instruction and intervention (Shogren et al., 2019). Understanding its influence on subjective well-being outcomes is therefore of interest in developing strategies for optimal well-being in the transition to adulthood.

Data Collection Instruments and Process

The survey questions consist of basic demographic characteristics, circumstances of education and employment in the present, educational attainment and future aspirations, questions concerning friendships, loneliness, social networks, social and political participation, language use in a range of contexts, and perceptions of identity. Many of these questions were modifications of those asked in other longitudinal general population studies of young people in the UK and Europe enabling the potential for comparison with general population studies. The survey was designed to be repeated annually with questions undergoing modifications (e.g., some would no longer be relevant depending on the age of the participant) and amendment. As issues emerged from the data analysis on a year-on-year basis, more probing in some highlighted areas might be justified, for example support in post-University employment. A core would remain the same to enable the analysis of pathways to adulthood on both individual and group bases. The baseline data set is archived through Figshare https://doi.org/10.48420/19566031 but data sharing is restricted until the end of the project.

All data collection takes place online and is managed through REDCap; a secure online digital data capture platform (Harris et al., 2009). Highly diverse language use is a feature of this generation of DHH young people, including in some cases fluid language use, increasingly referred to as translanguaging: the use of a personalized resource of language and communication that might draw on a range of modalities and languages for both comprehension and expression (Iturriaga & Young, 2021; Kusters, Spotti, Tapio, & Swanwick, 2017; Swanwick, 2017). To ensure this fluidity is accommodated, survey-based data collection is offered in a range of languages (English, Welsh, BSL) and modalities (written/spoken/signed, including Sign Supported Spoken English and Sign Supported Spoken Welsh). In addition, the research team created some bespoke coding within REDCap that meant that a young person could switch between languages and modalities on a question-by-question basis (Young, Espinoza, Dodds, Rogers, & Giacoppo, 2021). This had not been managed in REDCap previously. It permits the young person to use their linguistic strengths and capacities to maximize comprehension, rather than having to follow a survey in only one language.

The annual survey is thus available in written English, written Welsh, BSL, and in addition Sign Supported Spoken English (SSSE) and Sign Supported Spoken Welsh (SSSW). Questions are delivered by video but have high-quality acoustic recording as well, meaning that someone whose written literacy may lag behind their listening ability and comprehension in the spoken word would not be disadvantaged in completing the survey. All renderings of the survey in languages other than written English are produced by qualified and registered interpreters and checked by independent monitors. Several errors were found in initial translations, and we report on some of these in (Young et al., 2021) as the errors related not just to language but rather to the visual rendering of the question alongside online representations of the questions (e.g., the written text might have a response scale from left to right, the interpreter might have signed the scale as running from right to left).

Data collection for the three core outcome measures is also completed online and on a repeat basis annually, available in different language versions from which the participant might choose. A participant does not have to choose to complete a standard instrument in the same language they have completed the survey, which is to say they may do the survey in written English and the assessments in BSL. Unlike the within survey, it is not possible to switch language versions whilst completing single assessments because each version is validated for a specific language.

Subjective well-being is assessed using the 7-item Short Warwick and Edinburgh Mental Well-being Scale (SWEMWBS) (Stewart-Brown et al., 2009) available in validated versions in written English, written Welsh, and BSL that have been tested for reliability on the target language populations and, in the case of BSL, validated on a population of deaf BSL users (Rogers, Dodds, Campbell, & Young, 2018). SWEMWBS is responsive to change at both individual and group levels (Shah, Cader, Andrews, McCabe, & Stewart-Brown, 2021) and addresses both mental and emotional well-being (how "good" somebody feels) and psychological functioning (how well somebody thinks they are functioning). The 7-item SWEMWBS in written English has a Flesch Reading Ease score of 84.7 and Flesch–Kincaid Grade Level of 2.6 meaning it is suitable for young readers between 5 and 7 years old. We note the reading age score for this and the other standard instruments in recognition of the fact that some deaf users of written English may not have age-appropriate literacy in that language (Mayer, 2007). We wanted to ensure that should participants do the assessments in English, the language level of the instruments would not de facto be a barrier to comprehension but we acknowledge that this of itself is no guarantee of full comprehension.

The Self-Determination Inventory: Student Report (SDI:SR) (Shogren et al., 2017; Shogren, Shaw, Raley, & Wehmeyer, 2018) is suitable for young adults (up to the age of 22 years) and has been validated in populations with and without disabilities and in multiple languages. It has been translated into ASL and undergone initial validity testing with Deaf ASL users (Garberoglio et al., 2022) but no investigation of its validity with DHH young people who do not take it in a signed language has yet been carried out. In written English, it has a Flesch Reading Ease score of 92.6 and a Flesch–Kincaid Grade Level of 2.3 meaning it is suitable for those with a basic reading level. Under agreement with the originators, the study has modified the SDI:SR for UK English and created a BSL translation following a process similar to that used previously for other standard instruments in BSL (Rogers et al., 2016, 2018). An initial validation study is being carried out

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of both the UK English version as completed by a population of DHH young people, and the BSL version, concurrently with their implementation in the study (this will be reported in future publications). No version in Welsh is available.

The EQ-5D-5L is a standardized measure of health-related quality of life, available and validated in over 150 languages (EuroQol 2019, p.5) including BSL and reliability tested on an adult deaf population of BSL users (Rogers et al., 2016). The EQ-5D-5L in written English has a Flesch Reading Ease Score of 65.3 and a Flesch–Kincaid Grade Level of 5.9 indicating that it is suitable for "young readers." Results for this assessment are not reported here.

The READY study also has a qualitative interview component, not reported here, that engages a subsample of 48 participants in semi-structured interviews, the focus of which alters each time. In the first wave, it focused on life under Covid-19, social networks, participation, socialization, and self-determination. A specially recruited and trained group of diverse DHH co-inquirers worked with the main research team to determine the content of interviews and carried out over half of the interviews. They also co-constructed the data analysis framework. We report these findings separately.

Study Inclusion Criteria

To be eligible for participation in the READY study, young people had to: Have permanent deafness at any level of hearing (including unilateral deafness) and at any age of onset; be age 16–19 years at point of first data collection; be resident in England, Scotland, or Wales at entry into the study; be able to give individual informed consent (i.e., have capacity in line with the research provisions of the Mental Capacity Act 1998 in England and Wales and the Incapacity (Scotland) Act 2000, part 5). For purposes of this study assent or supported consent were not sufficient. The exclusion criteria were: possessing characteristics outside of those required for eligibility for participation in the study; unable to communicate effectively in any language, whether written, spoken, or signed. We acknowledge that these criteria meant that although DHH young people with additional needs were not excluded, those with a severe degree of intellectual disability or severe language deprivation would be excluded. We use the term "additional needs" in line with the UK usage to imply disabilities or learning needs in addition to being a DHH person. The term additional disabilities is not used as it assumes that deafness per se is a disability.

Recruitment

Three main recruitment pathways were used to reach potential participants but, in all cases, the young person themselves had to make direct contact with the research team if they were interested and wanted more information. The study targeted: (i) DHH young people in their last year of compulsory education in schools, with information distributed by teachers and through advertisements within schools. DHH young people could contact the study at age 15 years, expressing a wish to take part but would not be screened for eligibility and consented into the study until age 16 years. (ii) Post-16 educational and training settings such as Further Education Colleges, university disability support services, careers advisory services, and audiology clinics were asked to distribute information about the study to DHH young people they had contact with. (iii) Recruitment via wider social and congregational networks. This included advertising the study via social media, deaf clubs, youth groups, deaf sports organizations, and charitable support services such as the networks of the funder (the National Deaf Children's Society). Information materials to attract young people to the study were in English, Welsh, SSSE, and BSL with more details available on the dedicated study website in these languages. Information was also available in other written/spoken languages on request and quick links to alert to this in a range of other languages were provided.

Participants were screened for eligibility through the study's online portal before receiving the Participant Information Sheet available in plain English, plain Welsh, BSL, SSSE, or SSSW, or other languages on request. Those who completed the consent form (also available in multiple languages/formats) progressed to a short questionnaire which collected basic information about their background for the purpose of later sample stratification and were then directed to the main data collection materials.

Data collection began in July 2019 with Wave 1 intending to finish in June 2020. However, the Covid-19 pandemic hit in early 2020 with the UK experiencing its first serious lockdown from March 2020 to June 2020. Recruitment and data collection during this time was paused because of concerns that some of the questions in the study might be especially upsetting during a period of isolation for DHH young people. Consequently, the Wave 1a sample ran from July 2019 to mid-March 2020, with the Wave 1b sample running from July 2020 to June 2021. Subsequent waves of data collection are staggered depending on initial entry into the study, with an invitation to participate again on the 1-year anniversary of first providing data. Future publications will present longitudinal outcomes across waves.

Ethics

The study received ethical approval from the University of Manchester, UK, review board.

Analysis

Data from the main annual survey are presented through descriptive statistics including percentages, arithmetic means, and standard deviations. Statistical significance was set at a probability of 0.05 or less with effect sizes (ES) calculated using Cohen's *d* when assessing the relative importance of any categorical variables in bivariate analyses. Association was investigated using Spearman correlation analysis and t-tests. Regression analyses were conducted using both Bayesian and Frequentist approaches.

The SDI:SR is reported as a raw score (maximum score = 99) and by mean. This is calculated as an overall score and a score on each of the three internal constructs tested within the instrument: Volitional, Agentic, and Action-Control. Each of the seven statements of the SWEMWBS is scored on a 5-point scale (1–5); therefore, the minimum raw score is 7 and the maximum 35. However, the raw scores require conversion based on a standard table (Stewart-Brown et al., 2009) for valid comparisons with other data sets to be made. Additionally, scores may also be analyzed according to established clinical cut-offs for depression and for anxiety (Shah et al., 2021). Data from the EQ-5D-5L are analyzed in three different ways following the recommendations of Euroqol (2019, p.5) but not reported in this paper.

Results

RQ1: What Are the Characteristics of the Baseline Cohort?

Figure 1 illustrates the recruitment and progression of participants arriving at a baseline READY study cohort of 163 young people.

All background variables are collected by self-report and are not objectively checked. Full details are available in Table 1. The basic demographic characteristics of the 163 participants in the cohort show a larger number of women than men; this is not

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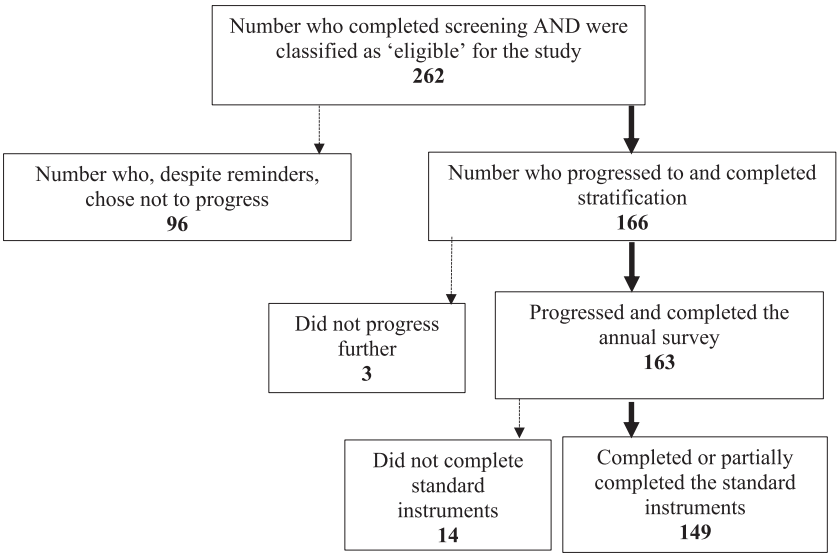

Figure 1. Ready study recruitment and participation.

unusual in cohorts of young adults recruited for research (Ryan et al., 2019). In terms of socioeconomic status, based on national indices of multiple deprivation in England, Scotland, and Wales (Ministry of Housing, Communities and Local Government, 2019; Scottish Government, 2020; Welsh Government, 2019) associated with specific postcodes (zip codes), the cohort has around a 6% over-representation of deaf young people from the least deprived quintiles, 4–5, and a 10% and 4% under representation in quintiles 1–2 respectively with the proportion in quintile 3 being close to expectation. Participants with severe/profound levels of deafness are over-represented in comparison with expected proportions for the DHH young adult population in England, Scotland, and Wales (CRIDE, 2019) but representation of those with moderate deafness and those who are unilaterally deaf is good. The vast majority (86.8%; $n = 142$) were born deaf or became deaf before the age of 5 years. Just over 9% ($n = 15$) came from a family where one or more parents was deaf; this proportion is in line with figures from other post-industrial nations (Mitchell & Karchmer, 2004). Fifty seven (37.2%) had at least one additional need which is a comparable reported incidence to other studies of deaf children/young people (Marschark, Hauser, Marschark, & Hauser, 2008). The proportion of the 163 recruited by country is in line with residence distributions of DHH young people by England, Wales, and Scotland (CRIDE, 2019). Ethnicity rather than race is routinely collected in national statistics in the UK, with seven distinctions of "White" made, four under "Asian", three under "Black", one under Arab, and three more addressing further classifications which we followed. In summarizing, we use the contested but common binary classification of "White" and "BAME" (Black, Asian, Minority Ethnic) in reporting our data because we are comparing with Office of National Statistics data; however subsequently the UK government recommended that BAME should not be used as a term in official statistical reporting going forward (Gov UK, 2021). 22.1% who identified as BAME is more than in the general population (12.8%) (NOMIS, 2021). Unusually for a sample of young DHH people, 49 (30.8%) identified as LGBTQ+ (Lesbian, Gay, Bisexual, Transgender, Queer or Questioning and others including pansexual and Two-Spirit) mirroring trends in the general population where prevalence of young people identifying as LGBTQ+ is higher than previously, and higher in comparison with older age groups (ONS, 2022).

The language profile of the READY cohort is complex. Participants, when asked, overwhelmingly stated that their preferred language(s) was a spoken language, with only 3 of the 163 saying they did not use spoken language at all. However, 116 (72.05%) said they knew some BSL and of the 159 who rated their proficiency in BSL, 36 (22.64%) rated it as very good or excellent. Further articles will report in depth on the language and communication profiles and habits of the cohort, who display a high degree of linguistic fluidity increasingly associated with translanguaging amongst deaf young people (Swanwick, 2017).

At baseline, over 94% were being educated in mainstream educational environments, only 7 were in work without any educational or training component associated with their employment (not apprenticeships or college with employment courses), and three were NEET (Not in Education, Employment or Training). Data were available on the educational qualifications of 129 participants (in 6 cases data were missing, in 28, participants had not yet had the opportunity to take a qualification and gain a result). Of the 129, 6 had no qualifications. Of the 123 who did, 115 (93%) had a level 2 qualification or higher. This corresponds to the highest level of qualification at first point of school leaving (aged 16 years). National statistics show that 73% of DHH young people will gain 5 or more level 2 qualifications by the age of 19 compared with 88% in the general population (Guardian, August 2019). In the READY cohort, 93% have done so by the same benchmark. The study will track the educational attainment and employment trajectories of the cohort over the next 5 years.

The following analyses presents results from participants who completed the standardized assessments in their written English form as this was the majority, and numbers who completed them in BSL were too low to make meaningful comparisons. The number of participants in each analysis varies given that not all participants completed all standard instruments.

RQ2: How Do Participants Score on the Outcome Measures of Subjective Well-Being and Self-Determination in Comparison with General Population Data?

National validation data on the SWEMWBS set an expected distribution of 15% in the low well-being category, 70% in medium well-being, and 15% in high well-being. Only four

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Table 1. READY baseline cohort demographic profile (N = 163).

Demographic characteristic	*n*	%
Sex (*n* = 163)		
Male	38	23.3
Female	123	75.5
Other	2	1.2
Gender identity descriptors (*n* = 162)		
Woman	118	72.8
Man	38	23.5
Transman	2	1.2
Non-binary	4	2.5
Sexuality (*n* = 159)		
Heterosexual	110	69.2
Gay	3	1.9
Lesbian	7	4.4
Bisexual	29	18.2
Other	10	6.3
Age at first enrolment (*n* = 163)		
15	17	10.4
16	52	31.9
17	51	31.3
18	32	19.6
19	11	6.8
Country of residence (*n* = 163)		
England	136	83.4
Wales	8	4.9
Scotland	19	11.7
Ethnicity (*n* = 163)		
White	127	77.9
BAME	36	22.1
Socioeconomic status (*n* = 150)		
Quintile 1 (most deprived)	16	10.7
Quintile 2	25	16.7
Quintile 3	29	19.3
Quintile 4	41	27.3
Quintile 5 (least deprived)	39	26.0
Degree of deafness (*n* = 163)		
Mild	3	1.8
Moderate	47	28.8
Severe	35	21.5
Profound	74	45.4
Don't know	4	2.5
Laterality of deafness (*n* = 163)		
Unilateral	23	14.1
Bilateral	140	85.6
Degree of deafness (bilateral only) (*n* = 140)		
Mild	3	2.1
Moderate	39	27.9
Severe	30	21.4
Profound	65	46.4
Don't know	3	2.1
Parents deaf (*n* = 155)		
Yes	15	9.2
No	148	90.8
Siblings deaf (*n* = 163)		
Yes	24	14.7
No	139	85.3
Deafness in family (*n* = 163)		
Parents deaf, no deaf siblings	9	5.5
Parents deaf, deaf siblings	6	3.7
Parents hearing, no deaf siblings	130	79.8
Parents hearing, deaf siblings	18	11.0
Age when became deaf (*n* = 163)		
Born deaf	98	60.1
0–5 years	44	26.7
5–11 years	10	6.1
11–16 years	11	6.8

(continued)

Table 1. Continued.

Demographic characteristic	*n*	%
Additional needs/disability (*n* = 153)		
At least one additional need or disability	57	37.2
No additional needs	96	62.8
Type of additional need/disability (n = 57) [1]		
Blind/visual impairment	24	42.1
Physical/motor difficulties	21	36.8
Learning difficulties	18	31.6
Mental health difficulties	3	5.3
Other	3	5.3
Amplification (*n* = 163)[a]		
Hearing aids	105	64.4
Bone conduction hearing device	6	3.7
Cochlear implants	47	28.8
None of these	7	4.3
Hearing aids (*n* = 105)		
One hearing aid	23	21.5
Two hearing aids	84	78.5
Cochlear implants (*n* = 47)		
One cochlear implant	19	38.0
Two cochlear implants	31	62
Educational setting (*n* = 145)[a]		
Local mainstream school	50	34.5
Mainstream school with a support base for deaf learners	14	9.7
Specialist deaf school	5	3.4
Special school for children with a range of difficulties	3	2.1
Sixth form linked to a mainstream school	24	16.6
Mainstream sixth form college	30	20.7
Specialist deaf college	1	.7
Special college for students with a range of difficulties	1	.7
FE college	13	9.0
University	14	9.7
Online courses	4	2.7
In a place that is not listed	9	6.2
Highest qualifications (*n* = 123)		
Entry level	2	1.6
Entry-level functional skills qualification		
Level 1	6	4.9
GCSE Grade D-G or Grade 3–1		
Level 2	80	65.0
GCSE/s Grade A∗ to C or Grade 9–4/Scottish National 5		
Level 3	34	27.6
A level/s/Scottish Higher/s		
Level 4	1	0.8
NVQ4		
Employment		
Employment only	7	
Employment and education	8	
NEET (not in employment, education, or training)	3	

participants in the READY cohort (3%) recorded high well-being (Figure 2). Using instead recommended clinical cut-off scores (Shah et al., 2021), 27 (20.4%) had scores of 18 or less indicating probable depression, 29 (22%) had scores between 18 and 20 indicating possible depression, and 76 (57.6%) had scores > 20 in the average or high well-being groups (treated as one category in the clinical validation study).

We compared the mean SWEMWBS scores of the group who completed the measure prior to the first major UK lockdown during the pandemic ($n = 77$, mean = 21.47, SD = 3.69) versus the group who completed the measure following the lifting of initial lockdown restrictions ($n = 55$, mean = 21.28, SD = 4.19). No statistically significant difference was found ($p = .763$).

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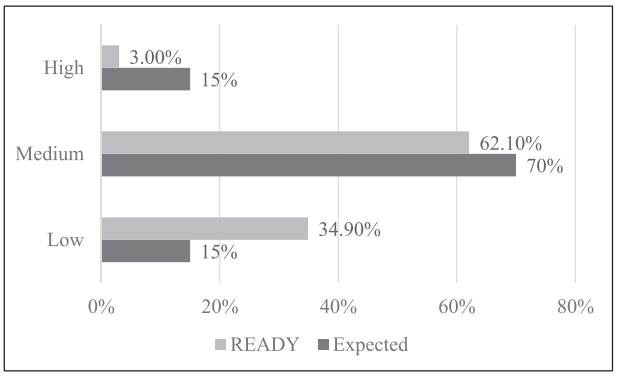

Figure 2. Expected versus READY distribution of well-being scores by category (written English only, $n = 132$).

The READY baseline cohort scored lower on the SDI:SR overall and in each domain (construct) than in the original SDI:SR validation study (Shogren et al., 2017) and lower than those "with disabilities" and those "without disabilities" in that study (Shogren et al., 2017) as well as lower than in a small study of DHH young people with a range of communication and language preferences in the USA (Millen, Dorn, & Luckner, 2019). Unlike results from an initial validation study of the SDI:SR with Deaf ASL users (Garberoglio et al., 2022), the READY cohort scores are not similar to those of youth without disabilities (Table 2).

We compared the mean SDI:SR scores of the group who completed the measure prior to the first major UK lockdown during the pandemic ($n = 79$, mean = 68.85, SD = 14.61) versus the group who completed the measure following the lifting of initial lockdown restrictions ($n = 55$, mean = 71.16, SD = 13.48). No statistically significant difference was found ($p = .355$).

RQ3: What Are the Relative Contributions of Sociodemographic Characteristics to Participants' Outcome Scores on the SWEMWBS and SDI:SR?

The mean score on the SWEMWBS is statistically significantly lower for women in the READY cohort than those in the general population age 16–24 group (Ng Fat, Scholes, Boniface, Mindell, & Stewart-Brown, 2017), $p < .0001$, ES: 0.59. A similar pattern appears evident for men, but the study is underpowered with respect to male participants for a *t*-test to detect statistical difference.

The relationship between having additional needs and well-being scores was examined by comparing the mean scores of those without additional needs ($n = 80$, mean 21.6, SD = 3.7, 95% CI 20.7822.4) and those with at least one additional need ($n = 45$, mean 21.2, SD = 3.8, 95% CI 19.9922.3). The means are very similar and there are overlapping SDs. Therefore, although sample size does not permit a meaningful test of statistical difference (ES: 0.12), analysis suggests that having additional needs does not appear to influence subjective well-being. There was no difference in subjective well-being scores between those of a White ethnicity ($n = 100$, mean 21.4, SD = 3.6, 95% CI 20.622.1) compared to BAME ($n = 32$, mean 21.2, SD = 4.7, 95% CI 19.5–22.9); ES: 0.03 (no effect). To explore the relationship between degree of deafness and SWEMWBS scores, participants were regrouped into two bands: lower degree of deafness (unilateral, mild, and moderate, $n = 54$) and higher degree of deafness (severe, profound, $n = 77$). The group in the lower degree of deafness band have a mean SWEMWBS score of 20.9 ($n = 54$, SD = 4.17, 95% CI 18.77–22.04), whereas the group in the higher degree of deafness band have a mean SWEMWBS score of 21.6 ($n = 77$, SD = 3.71, 95% CI 20.78–22.47). The small effect size (ES: 0.18) precludes statistical testing for significance but given the closeness of the means we conclude that degree of deafness does not exert a discernible effect on SWEMWBS scores for the READY sample. A marginal difference was identified (ES: 0.23) between those in the 50% most deprived areas/neighborhoods (as per official figures of Indices of Multiple Deprivation in each constituent country) and 50% living in the least deprived areas, with those in the least deprived having a slightly higher mean score of 21.75. However, using a two-sample *t*-test with equal variances, differences in SWEMWBS scores are statistically significant between participants who are LGBTQ+ ($n = 40$, mean 19.6, SD = 4.2, 95% CI 18.2–20.9) and those who are heterosexual/straight ($n = 90$, mean 22.1, SD = 3.5, 95% CI 21.38–22.84) participants (ES: 0.68; $p < .001$) (Table 3).

No significant differences in SDI:SR score were observed according to age, ethnicity, degree of deafness, additional needs, SES, or sex (Table 4). However, LGBTQ+ participants have lower scores in every dimension of the SDI:SR and overall, in comparison with heterosexual/straight participants, and differences in overall mean SDI:SR scores show a large effect size and are significant ($p < .001$, ES: 0.87).

RQ4: To What Extent Does Self-Determination Influence Subjective Well-Being?

Although these descriptive analyses of group average self-determination and well-being indicated low levels of scoring for the group as a whole in comparison with general population data for the same age group, a cursory view of individual data seemed to suggest that those who scored highly on self-determination also scored highly on well-being (Figure 3). We, therefore, next examined whether these two variables are positively associated using two tests of association; Pearson *r* (0.533, $p < .001$) that assumes a linear association and Spearman rho (0.533, $p < .001$) which considers values as ranked and is more flexible for non-linear relations. We conclude there is a positive association.

A multiple regression analysis determined the relative contribution of the sociodemographic background variables to the SWEMWBS scores separate from the potential contribution of SDI:SR scores as a predictor variable. A Bayesian approach was preferred because it works better with smaller samples and its requirements have a degree of flexibility. Sociodemographic predictors of well-being included are sex (men, women), sexual orientation/gender identity (heterosexual, LGBTQ+), socioeconomic status (quintiles based on residential postcode), degree of deafness (CRIDE scale), having at least one additional need, and level of educational attainment (highest qualification). Overall, this model explains 11.6% (R2 Bayes = 0.116) of the result, which is a small proportion of the variation in well-being scores. We conclude none of the sociodemographic characteristics exert a significant effect on the SWEMWBS results (Table 5).

Next, scores on the SDI:SR for self-determination were added as a predicator variable. This increases the explanatory power of the model to 37.7% (R2 Bayes = 0.377) which is considered good for analysis from a cross-sectional study. In this model, well-being scores increase by 0.18 per each additional point on the self-determination index. We conclude: (i) sociodemographic variables explain very little of the variance in SWEMWBS scores; (ii) higher levels of self-determination are a predictor of higher levels of well-being, outweighing the influence of any background characteristics; (iii) although women and those who are LGBTQ+ have statistically significantly lower well-being scores (as previously demonstrated), these aspects of their identity cannot of themselves be seen as predictive risk factors.

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Table 2. READY baseline cohort SDI:SR scores (written English only, $n = 125$) in comparison with other studies.

Research study	(Shogren et al., 2017) with disabilities ($n = 176$)	Shogren et al., 2017) without disabilities ($n = 135$)	Millen et al., 2019 ($n = 29$)	READY baseline cohort ($n = 125$)		
	M	m	m	SD	m	SD
Construct						
Volitional action	75	83	-	-	69.77	14.63
Agentic action	74	83	-	-	70.16	15.80
Action control beliefs and attitudes	77	81	-	-	70.85	16.3
Overall average score	75.3	82.3	75.4	13.8	70.18	14.40

Table 3. SWEMWBS scores (written English only) demographic variables ($n = 132$).

Demographic characteristic	n	M	SD	95% CI	p	Cohen's d
Ethnicity					.835	0.03
White	100	21.37	3.64	2.64–22.09		
BAME	32	21.22	4.65	19.54–22.90		
Additional needs					.519	0.12
No additional needs	80	21.60	3.67	10.78–22.41		
Have additional needs	45	21.15	3.83	19.99–22.30		
Deafness					.299	0.18
Lesser severity grouping	54	20.91	4.17	18.77–22.04		
Higher severity grouping	77	21.63	3.71	20.78–22.47		
Sexuality					.0005	0.68
LGBTQ+	40	19.56	4.24	18.20–20.921		
Heterosexual/Straight	90	22.11	3.49	21.38–22.84		
SES					.222	0.23
50% most deprived	44	20.83	4.68	19.40–22.25		
50% least deprived	76	21.75	3.50	20.95–22.58		
Lockdown					.893	0.02
Pre-lockdown[a]	77	21.42	3.69	20.58–22.22		
Post-lockdown	55	21.21	4.19	20.08–22.34		

[a]The cut off for pre-first UK lockdown was March 23, 2020. Data collection resumed on June 30, 2020.

Table 4. SDI:SR scores (written English only) demographic variables ($n = 133$).

Demographic characteristic	N	M	SD	95% CI	p	Cohen's d
Sex					.477	0.146
Men	31	71.61	14.35	66.34–76.88		
Women	102	69.57	13.81	66.85–72.28		
Ethnicity					.486	0.139
White	33	68.30	12.91	63.72–72.88		
BAME	101	70.28	14.56	67.41–73.16		
Additional needs					.061	0.346
No additional needs	80	71.85	12.27	69.02–74.67		
Have additional needs	47	67.16		62.82–71.51		
Deafness					.248	0.203
Lesser severity grouping	56	68.08	14.24	64.27–71.90		
Higher severity grouping	77	70.97	14.13	67.76–74.18		
Sexuality					<.001	0.871
LGBTQ+	40	61.77	15.56	56.79–66.75		
Heterosexual/straight	92	73.25	11.7	70.79–75.71		
SES					.754	0.053
50% most deprived	45	69.56	15.18	65.00–74.12		
50% least deprived	77	70.38	13.11	67.40–73.35		
Lockdown					.254	0.202
Pre-lockdown[a]	78	68.59	14.52	65.32–71.86		
Post-lockdown	56	71.48	13.56	67.84–75.11		

[a]The cut off for pre-first UK lockdown was March 23, 2020. Data collection resumed on June 30, 2020.

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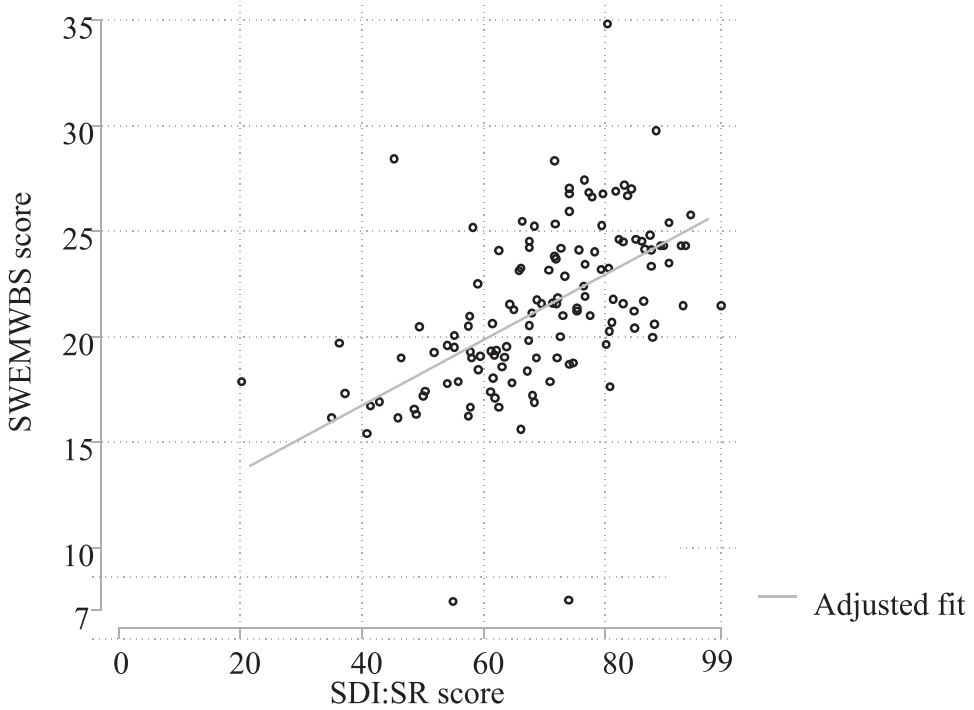

Figure 3. Association of SDI:SR scores with SWEMWBS scores (written English only).

Discussion

In summary, this is an educationally high achieving cohort, who are over-represented in the severe and profound categories of deafness and the higher socioeconomic groupings, with a greater representation of young people coming from a range of Black, Asian, and Minority Ethnic backgrounds than in the general population. Prevalence of additional needs within this cohort is comparable with general data on DHH young people and the proportion of those who have one or more parent who is also DHH is as expected. The predominance of spoken language as first and preferred language of 95% of the cohort is offset by nearly three quarters having some BSL skills, with around one quarter rating their proficiency in BSL as very good or excellent. The percentage of those identifying as LGBTQ+ mirrors trends identified in the general population, where higher prevalence is recorded for this age group (Generation Z) than for other ages and generations (ONS, 2022). The vast majority of the READY cohort at baseline are age 16 and 17 and still living at home with parents and other family members.

We reiterate these characteristics as we are not claiming that the READY cohort is statistically representative of the full range of characteristics and backgrounds of deaf young people. We are claiming that it is a good representation of a range of possibilities, trends, and social circumstances that are common to Generation Z deaf young people. The testing of the contribution of a range of sociodemographic features on outcomes measured is fundamental to understanding this cohort and influences on their generation, along with other contextual features of being deaf in the 21st century that are captured in other aspects of this study, separately reported. The focus on self-determination and subjective well-being as one such important association that is open to testing over time as young people mature and their circumstances change. Importantly, in trying to understand risk and protective factors for successful maturity in young adulthood, self-determination is a dispositional trait that is open to change and development and therefore potentially to successful intervention (Shogren et al., 2019).

The READY cohort scores on subjective well-being are a cause for concern with only 3% registering high well-being in comparison with an expected level of 15% in a general population sample of the same age. The 42% with probable or possible depression is less than that found in the study of 11–16 year olds by Roberts et al. (2015) in the UK who identified 56 out of 79 (71%) DHH young people with a probable psychiatric disorder during a validation of the Strengths and Difficulties Questionnaire (SDQ) in BSL, and is nearer the 43% (34 out of 79) identified by clinical interview alone. The higher (better) scores for men in comparison with women in the sample is in line with trends commonly seen in studies of well-being in young people of this age in the general population (Ng Fat et al., 2017). Although based on a sample of DHH adults older than the READY cohort, data derived from the HINTS—ASL (Health Information National Trends Survey—American Sign Language) have observed something similar, demonstrating that deaf ASL users who are White, younger, female, educated, or single were significantly more likely to have a diagnosis of depression/anxiety based on health care provider patient records (Kushalnager, Reesman, Holcomb, & Ryan, 2019; Kushalnagar et al., 2020).

However, further analysis of the relative contribution of a range of sociodemographic features to subjective well-being outcomes has demonstrated that although women and those who are LGBTQ+ have significantly lower well-being scores both within the cohort and in comparison with general population norms, these aspects of their identity cannot of themselves be seen as predictive risk factors. The influence of self-determination on subjective well-being outweighs that of any other background variable, including sex and sexual orientation.

That said, the consistently significantly lower scores from LGBTQ+ participants on the outcome measures of self-determination and subjective well-being is important to consider, as not all young people will face the same challenges in well-being

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Table 5. Contribution of sociodemographic and self-determination scores (SDI:SR) to well-being scores (SWEMWBS), READY Study baseline cohort (written English only), $n = 105$.

	Sociodemographic predictor variables only		Sociodemographic predictor variables + SDI:SR scores as predictor variable	
Predictors	**Estimates**	**CI (95%)**	**Estimates**	**CI (95%)**
Intercept	21.07	17.54 to 24.58	11.29	7.15 to 15.41
Sex: female	−1.77	−3.68 to 0.13	−1.16	−2.74 to 0.44
Sexual orientation: heterosexual	1.49	−0.25 to 3.23	0.18	−1.31 to 1.68
Socioeconomic status (quintile deprivation)	0.10	−0.48 to 0.69	−0.29	−0.80 to 0.21
Degree of deafness	0.12	−0.48 to 0.72	−0.08	−0.57 to 0.43
Have additional needs	−0.01	−1.59 to 1.57	0.48	−0.87 to 1.82
Qualifications level	0.02	−0.51 to 0.55	−0.08	−0.52 to 0.36
SDI:SR scores	-	-	0.18	0.12 to 0.23
Observations	105		105	
R^2 Bayes	0.116		0.377	

through late adolescence. Very little data exist on the health and well-being of LGBTQ+ DHH people of any age. However, sexuality and gender identity has been demonstrated to impact on DHH people's health. Miller, Biskupiak, and Kushalnagar's (2019) study of interactions with health care providers evidences that, after controlling for sociodemographic and patient-related variables, DHH cisgender women were significantly less likely to disclose their LGBTQ+ status compared with DHH cisgender men. Sanfacon et al. (2021) focusing on a sample of 74 transgender Deaf adults who use ASL, identified a lifetime prevalence of 48.6% for depression/anxiety disorder and further concluded that identification as non-binary increased a Deaf person's risk for being diagnosed with depression or anxiety disorder by 80% (95% confidence interval, 1.11–2.90) relative to Deaf people who self-identified as a binary gender. The READY cohort will record both changing and stable identifications of sexual orientation, gender identity, deaf identity, and language use over time in a way that previous studies have been unable to and will relate these to outcome measures (Tables 2–5).

Finally, analysis so far strongly supports the possibility of improving well-being through enhancing self-determination. Well-being scores increase by 0.18 per each additional point on the self-determination index. Self-determination is a dispositional trait open to modification, particularly up to the age of 25 years. These results point to the importance of work being carried out to develop self-determination interventions with young people with disabilities (Burke et al., 2020; Raley, Mumbardó-Adam, Shogren, Simó-Pinatella, & Giné, 2018). The Self Determined Learning Model of Instruction (SDLMI) which is classroom based is widely disseminated and recommended through the National Deaf Center for Postsecondary Outcomes in the USA (Shogren et al., 2019). The evidence base for its effectiveness shows some promising early results of its impact on the ability of DHH students to set and attain goals at a better level of progress than expected (Spolsky, 2014) but definitive evidence of effectiveness for DHH young people is yet to be established.

Limitations

This is a self-selecting sample that are not necessarily representative of average educational attainment and proportionate socioeconomic status, and which precludes the most severely disabled, intellectually challenged and language deprived groups within the population of DHH young people. The small number who did complete in BSL ($n = 10$) made meaningful comparison infeasible. Those who completed the outcome measures in written English had alternative choices (including Sign Supported Spoken English and BSL); therefore, it is reasonable to conclude they made a choice, rather than being forced to respond by default in written English. Also, the reading age requirements for all the instruments was low enough to ensure reading age in English was not an unnecessary barrier. However, reading age is not necessarily a good indicator of conceptual understanding or world knowledge, which can be challenging for some DHH young people, arising from limited exposure to a diversity of information through inter-subjective relationships with a diverse group of interlocutors, sometimes termed "fund of information deficit" (Pollard, 1998). The use of educational attainment as a generalized proxy for level of literacy could also be criticized, as there is no objective measure of literacy. Although DHH young people completed the outcomes assessments in written English, for which they are validated, there is no data on their reliability and internal or construct validity when taken by a population of young people who are DHH. The general population data with which comparisons have been made with response to the scores of the READY cohort contains samples of young people classified as disabled, some of which include deaf young people. Therefore, our comparative population data is not with hearing people, but with general population normative data. Self-report rather than objective measures of deafness could also be questioned as reducing the robustness of our analysis by levels of deafness; however, the fact that degree of deafness does not appear to excerpt any major influences on outcomes measures suggests that this may not be of concern. The exclusively online nature of the data collection may be regarded as exerting some unintended exclusions because of digital poverty reducing the possibility of online participation for some people, which we accept.

Conclusion

The READY study will potentially fill an important knowledge gap in being an exclusively self-report prospective longitudinal study of DHH young people from Generation Z. Its achievement of a baseline of 163 will mean that, even with inevitable drop out, the sample will be large enough to meaningfully provide repeat data over time on key outcome measures of health, well-being, and self-determination whilst plotting individual and group trajectories of the influence of socioeconomic status, educational attainment, social networks, and social participation through crucial transition years. The cohort are representative of a group

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of DHH entering adulthood with good educational attainment, who are predominantly spoken language users exploring also signed language, and who have given us the opportunity to look more deeply into their everyday lives and how these are changing over time. Our initial results on subjective well-being point clearly to the importance of the development of individual young people's self-determination as both a preventative measure to avoid clinical consequences of low well-being and as a positively predictive intervention to support the further enhancement of well-being.

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